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EDUCATION	**2025 Doctor of Biological Sciences (DSc), Genetics (1.5.7)** Research Centre for Medical Genetics, Moscow, Russia **2010 Candidate of Biological Sciences (PhD), Molecular Genetics (03.01.07)** Institute of Chemical Biology and Fundamental Medicine, Siberian Branch of the Russian Academy of Sciences, Novosibirsk, Russia **2006 Degree in Biology, specialization in Genetics** Chelyabinsk State University, Chelyabinsk, Russia
CURRENT PROFESSIONAL APPOINTMENTS	**2025 - present: Chief Executive Officer, MODICUM LLC,** Moscow, Russia **2024 - present: Head of Department of Biology and Genetics,** Petrovsky Medical University, Moscow, Russia **2024 - present: Leading Research Scientist,** Laboratory of Experimental Gene Therapy for Inherited Metabolic Diseases, Research Centre for Medical Genetics named after Academician N.P. Bochkov, Moscow, Russia **2019 - present: Senior Research Scientist,** Center for Bio- and Medical Technologies, Skolkovo Institute of Science and Technology (Skoltech), Moscow, Russia
ADDITIONAL ACTIVITIES	**2026 - present: Co-founder, CellKinetica SARL (Switzerland),** a company that uses genomic data analysis to provide additional information for ranking human embryos for transfer in assisted reproductive technology cycles. **2025 - present: Co-founder, Florence Labs LLC,** a biotechnology company developing reagents and software for whole-genome screening of embryos in assisted reproductive technology cycles. **2025 - present: Founder, MODICUM LLC, Skolkovo resident,** an IT company that uses genomic data analysis to provide additional information for ranking human embryos for transfer in assisted reproductive technology cycles. **2021 - present: Board Member,** Interdisciplinary Clinical Association for Reproductive Medicine (ICARM) https://www.mkarm.org/eng
PAST PROFESSIONAL APPOINTMENTS	**2024 - 2026: Deputy Head,** Reference Center for Assessment of Genetic Purity of Laboratory Animals, Avtsyn Research Institute of Human Morphology, Moscow, Russia **2020 - 2023: Head of Genetic Laboratory,** Medical Genomics LLC, Moscow and Tver, Russia **2020 - 2022: Scientific Director,** Network of Medical Centers "Fomin Clinics", Russia **2019 - 2020: Leading Research Scientist,** Genome Research Centre, Immanuel Kant Baltic Federal University, Kaliningrad, Russia **2012 - 2019: Head of Molecular Genetics Laboratory,** Immanuel Kant Baltic Federal University, Kaliningrad, Russia

	2016 - 2019: Associate Professor, Institute of Living Systems, Immanuel Kant Baltic Federal University, Kaliningrad, Russia **2012 - 2015: Associate Professor,** Department of Obstetrics and Gynecology, Medical Institute, Immanuel Kant Baltic Federal University, Kaliningrad, Russia **2011 - 2012: Research Scientist,** Ural Research Center for Radiation Medicine, Chelyabinsk, Russia **2009 - 2011: Junior Research Scientist,** Institute of Chemical Biology and Fundamental Medicine, Siberian Branch of the Russian Academy of Sciences, Novosibirsk, Russia **2006 - 2009: Senior Laboratory Assistant,** Institute of Cytology and Genetics, Siberian Branch of the Russian Academy of Sciences, Novosibirsk, Russia
TEACHING MANUALS	*Cell Biology* [Electronic resource]: textbook for medical students / I.O. Mazunin, Yu.V. Markina, T.V. Kirichenko, B.A. Rimskaya, A.M. Markin. Petrovsky Medical University, Moscow. Electronic data (16.3 MB). Moscow: MDV Group, 2026. ISBN 978-5-906748-37-9 *Template Processes in the Cell* [Electronic resource]: textbook for medical students / I.O. Mazunin, T.V. Kirichenko, Yu.V. Markina, I.A. Marinina, B.A. Rimskaya, U.S. Khovantseva. Petrovsky Medical University, Moscow. Electronic data (15.8 MB). Moscow: MDV Group, 2026. ISBN 978-5-906748-38-6
PUBLICATIONS (ORIGINAL RESEARCH)	Tofilo M., Volodyaev I., Tretiakov E., Kokorina O., Avdeichik S., Smirnova A., Grechenko E., Filimonova A., Sudarikova N., Kirillova A., Mazunin I. *Concordant mtDNA heteroplasmy between trophectoderm and inner cell mass supports blastocyst-stage PGT-mt for m.14487T>C* ***Journal of Assisted Reproduction and Genetics 2026*** (doi: in press) Pilicheva M., Rimskaya B., Fomich M., Povedailo V., Mazunin I., Shmanai V. *Synthesis of spectrally tunable cyanine analogs of mitotracker deep red for mitochondrial imaging* ***Bioorganic Chemistry 2026*** (doi: 10.1016/j.bioorg.2026.110162) Rimskaya B., Kropocheva E., Ponomareva I., Karchemkina L., Lisitskaya L., Gelfenbein D., Ulashchik E., Shmanai V., Kulbachinskiy A., Mazunin I. *Activity of DNA- and RNA-Guided Prokaryotic Argonautes in Human Mitochondria* ***Cells 2026*** (doi: 10.3390/cells15121129) Zakirova E., Sergeeva S., Morozova K., Kiseleva E., Tanaka M., Mazunin I., Orishchenko K. *Specific elimination of m.8993T>G mitochondrial haplotype in NARP cybrid cells by CRISPR-Cas9 system* ***Scientific Reports 2026*** (doi: 10.1038/s41598-026-49007-y) Rimskaya B., Kropocheva E., Shchukina E., Ulashchik E., Gelfenbein D., Lisitskaya L., Shmanai V., Smirnikhina S., Kulbachinskiy A., Mazunin I. *Targeting of human mitochondrial DNA with programmable pAgo nuclease* ***Cells 2026*** (doi: 10.3390/cells15020127) Nikitchina N., Hecke A-M., Shebanov N., Mazunin I., Tarassov I., Entelis N. *Targeted deletions in human mitochondrial DNA engineered by Type V CRISPR-Cas12a system* ***NAR Molecular Medicine 2025*** (doi: 10.1093/narmme/ugaf021)

Iliushchenko D., Efimenko B., Mikhailova A.G., Shamanskiy V., Saparbaev M.K., Matkarimov B.T., Mazunin I., Knorre D., Kunz W.S., Denisov S., Khrapko K., Fellay J., Gunbin K., Popadin K.
Deciphering the Foundations of Mitochondrial Mutational Spectra: Replication-Driven and Damage-Induced Signatures Across Chordate Classes
Molecular Biology and Evolution 2025
(doi: 10.1093/molbev/msae261)

Fan F., Povedaiko V.A., Lysenko I.L., Sharko O.L., Mazunin,I.O., Tikhomirov S.A., Shmanai V.V.
Photophysical features of protolytic equilibria of fluorescein bifluorophores
Journal of Applied Spectroscopy 2024
(V. 91, N 5, PP 723-32)

Belyaev A., Tofilo M., Popov S., Mazunin I., Fomin D.
The Assessment of Methods for Preimplantation Genetic Testing for Aneuploidies Using a Universal Parameter: Implications for Costs and Mosaicism Detection
Journal of Biotechnology and Biomedicine 2024
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Nikitchina N., Ulashchik E., Shmanai V., Hecke I A-M., Tarassov I., Mazunin I., Entelis N.
Targeting of CRISPR-Cas12a crRNAs into human mitochondria
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Fan F., Povedailo V.A., Lysenko I.L., Seviarynchyk T.P., Sharko O.L., Mazunin I.O., Shmanai V.V.
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Live birth of a healthy child in a couple with identical mtDNA carrying a pathogenic c.471_477delTTTAAAAinsG variant in the MOCS2 gene
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Shamanskiy V., Mikhailova A.A, Tretiavov E.O., Ushakova K., Mikhailova A.G., Oreshkov S., Knorre D., Ri N., Overdevest J.B., Lukowski C-W., Gostimskaya I., Yurov V., Liuo S.W.,, Lin T-K., Kunz W., Reymond A., Mazunin I., Bazykin G.A., Fellay J., Tanaka M., Khrapko K., Gunbin K., Popadin K.
Secondary structure of the human mitochondrial genome affects formation of deletions
BMC Biology 2023
(doi: 10.1186/s12915-023-01606-1)

Mikhaylova A.G., Mikhailova A.A., Ushakova K., Tretiakov E.O., Illushchenko D., Shamansky V., Lobanova V., Kozenkov I., Efimenko B., Yurchenko A.A., Kozenkova E., Zdobnov E., Makeev V., Yurov V., Tanaka M., Gostimskaya I., Fleischmann Z., Annis S., Franco M., Wasko K., Denisov S., Kunz W., Knorre D., Mazunin I., Nikolaev S., Fellay J., Reymond A., Khrapko K., Gunbin K., Popadin K.
A mitochondria-specific mutational signature of aging: increased rate of A>G substitutions on a heavy strand
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Vasilev R., Gunitseva N., Shebanova R., Korzhenkov A., Vlaskina A., Evteeva M., Polushkina I., Nikitchina N., Toshchakov S., Kamenski P., Patrushev M., Mazunin I.
Targeted modification of mammalian DNA by a novel Type V Cas12a endonuclease from Ruminococcus bromii
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Tofilo M., Voskoboeva E., Timonina V., Staufner C., Hoffmann G., Poschl J., Schlieben L., Prokisch H., Kholodov D., Ladygin S., Avdeichik S., Mazunin I.
A new family with congenital myasthenic syndrome caused by compound heterozygous CHAT mutations
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Shebanova R., Nikitchina N., Shebanov N., Mekler V., Kuznedelov K., Ulashchik E., Vasilev R., Sharko O., Shmanai V., Tarassov I., Severinov K., Entelis N., Mazunin I.
Efficient target cleavage by Type V Cas12a effector programmed with split CRISPR RNA
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A new family with Epiphyseal chondrodysplasia type Miura
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Mitochondrial DNA variation of Leber's Hereditary Optic Neuropathy (LHON) in Western Siberia
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PLOS One 2019
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Normal to Obesity and Back: The Associations between Mitochondrial DNA Copy Number, Gender, and Body Mass Index
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(doi: 10.3390/cells8050430)

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Interrelation of chemerin and TNF-α with mtDNA copy number in adipose tissues and blood cells in obese patients with and without type 2 diabetes
BMC Medical Genomics 2019
(doi: 10.1186/s12920-019-0485-8)

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The relationship between the mtDNA copy number in insulin-dependent tissues and markers of endothelial dysfunction and inflammation in obese patients
BMC Medical Genomics 2019
(doi: 10.1186/s12920-019- 0486-7)

Bykov R.A., Yudina M.A., Gruntenko N.E., Zakharov I.K., Voloshina M.A., Melashchenko E.S., Danilova M.V., Mazunin I.O., Ilinsky Y.Y.
Prevalence and genetic diversity of Wolbachia endosymbiont and mtDNA in Palearctic populations of Drosophila melanogaster
BMC Evolutionary Biology 2019
(doi: 10.1186/s12862-019-1372-9)

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Additional evidence of the link between mtDNA copy number and the body mass index
Mitochondrial DNA Part A: DNA Mapping, Sequencing, and Analysis 2018

Skuratovskaia D.A., Sofronova J.K., Zatolokin P.A., Vasilenko M.A., Litvinova L.S.,Mazunin I.O.
The association of the mitochondrial DNA OriB variants with metabolic syndrome
Biochemistry (Moscow) Supplement Series B: Biomedical Chemistry 2017

Sofronova J.K., Ilinsky Y.Y., Orishchenko K.E., Chupakhin E.G., Lunev E.A., Mazunin I.O.
Detection of mutations in mitochondrial DNA by droplet digital PCR
Biochemistry (Moscow) 2016

Orishchenko K.E., Sofronova J.K., Chupakhin E.G., Lunev E.A., Mazunin I.O.
Delivery Cas9 into mitochondria
Genes & Cells 2016

Toshchakov S.V., Korzhenkov A.A., Samarov N.I., Mazunin I.O., Mozhey O.I., Shmyr I.S., Derbikova K.S., Taranov E.A., Dominova I.N., Bonch-Osmolovskaya E.A., Patrushev M.V., Podosokorskaya O.A., Kublanov I.V.
Complete genome sequence of and proposal of Thermofilum uzonense sp. nov. a novel hyperthermophilic crenarchaeon and emended description of the genus Thermofilum
Standards in Genomic Sciences 2015

Sokhonevich N.A., Yurova K.A., Gutsol A.A., Khaziakhmatova O.G., Mazunin I.O., Litvinova L.S.
Effects of Immunoregulatory Cytokines (IL-2, IL-7, and IL-15) on Expression of Gfi1 and U2afll4 Genes in T Cells at Different Stages of Differentiation
Bulletin of Experimental Biology and Medicine 2015

Mozheĭ O.I., Zatolokin P.A., Vasilenko M.A., Litvinova L.S., Kirienkova E.V., Mazunin I.O.
Evaluating the mitochondrial dna copy number in leukocytes and adipocytes from metabolic syndrome patients: pilot study
Molecular Biology 2014

Litvinova L.S., Mazunin I.O., Gutsol A.A., Sokhonevich N.A., Khaziakhmatova O.G., Kofanova K.A.
Dose-response effect of steroid hormones on Gfi1 and U2af1l4 gene expression in T lymphocytes at different stages of differentiation
Molecular Biology 2013

Sukernik R.I., Volodko N.V., Mazunin I.O., Eltsov N.P., Dryomov S.V., Starikovskaya E.B.
Mitochondrial genome diversity in the tubalar, even, and ulchi: Contribution to prehistory of native siberians and their affinities to native americans
American Journal of Physical Anthropology 2012

Sukernik R.I., Volodko N.V., Mazunin I.O., Eltsov N.P., Starikovskaya E.B.
The genetic history of Russian old settlers of polar northeastern Siberia
Russian Journal of Genetics 2010

Eltsov N.P., Volodko N.V., Starikovskaya E.B., Mazunin I.O., Sukernik R.I.
The role of natural selection in the evolution of mitochondrial haplogroups in Northeastern Eurasia
Russian Journal of Genetics 2010

Volodko N.V., Starikovskaya E.B., Mazunin I.O., Eltsov N.P., Naidenko P.V., Wallace D.C., Sukernik R.I.
Mitochondrial Genome Diversity in Arctic Siberians, with Particular Reference to the Evolutionary History of Beringia and Pleistocenic Peopling of the Americas
American Journal of Human Genetics 2008

PUBLICATIONS (REVIEW ARTICLES)

Shchukina E., Eremin I., Mazunin I., Moskalev A.
Gene therapy methods and delivery approaches (review)
Modern Technologies in Medicine 2025
(doi: 10.17691/stm2025.17.6.06)

Shchukina E., Eremin I., Mazunin I., Moskalev A.
Mini Review: Gene Therapy Targets for Aging-Associated Diseases
Biochemical and Biophysical Research Communications 2025
(doi: 10.1016/j.bbrc.2025.152708)

Ivanova A., Korchivaya E., Semenova M., Lebedev I., Mazunin I., Volodyaev I.
The chromosomal challenge of human embryos: mechanisms and fundamentals
Human Genetics and Genomics Advances 2025
(doi: 10.1016/j.xhgg.2025.100437)

Volodyaev I., Ivanova A., Korchivaya E., Surnov A., Pomerantseva E., Lebedev I., Semenova M., Mazunin I.
The chromosomal challenge of human embryos: prevalence of aneuploidy and mosaicism
F&S Reviews 2025
(doi: 10.1016/j.xfnr.2024.100082)

Rimskaya B., Shebanov N., Entelis N., Mazunin I.
Enzymatic tools for mitochondrial genome manipulation
Biochimie 2025
(doi: 10.1016/j.biochi.2024.10.013)

Korchivaya E., Silaeva Y., Mazunin I., Volodyaev I.
The mitochondrial challenge: disorders and prevention strategies
BioSystems 2023
(doi: 10.1016/j.biosystems.2022.104819)

Kirillova A., Mazunin I.
Operation "mitochondrial wipeout" - clearing recipient mitochondria DNA during nuclear transfer
Journal of Assisted Reproduction and Genetics 2022
(doi: 10.1007/s10815-022-02561-6)

Kirillova A., Smitz J.E.J., Sukhikh G., Mazunin I.
The role of mitochondria in oocyte maturation
Cells 2021
(doi: 10.3390/cells10092484)

Zakirova E., Muzyka V., Mazunin I., Orishchenko K.
Natural and artificial mechanisms of mitochondrial genome elimination
Life 2021
(doi: 10.3390/life11020076)

Gazizova I.R., Mazunin I.O., Malishevskaya T.N., Kiseleva O.A., Gadzhiev A.M., Rindzhibal Al-M.
Mitochondrial DNA as a Factor of Glaucomous Optic Neuropathy's Development Mechanism
Oftalmologiya 2019
(doi: 10.18008/1816-5095-2019-4-479-486)

Verechshagina N., Nikitchina N., Yamada Y., Harashima H, Tanaka M., Orishchenko K., Mazunin I.
Future of human mitochondrial DNA editing technologies
Mitochondrial DNA Part A: DNA Mapping, Sequencing, and Analysis 2019

Mazunin I.
Human mitochondrial genome surgery
Genes & Cells 2018
(doi: 10.23868/201811030)

Verechshagina N.A., Konstantinov Y.M., Kamenski P.A., Mazunin I.O.
Protein and nucleic acid import into mitochondria
Biochemistry (Moscow) 2018

Mazunin I.O., Volodko, N.V.
Leber hereditary optic neuropathy
Vestnik Oftalmologii 2018

Mazunin I.O., Levitskii S.A., Patrushev M.V., Kamenski P.A.
Mitochondrial matrix processes
Biochemistry (Moscow) 2015

Patrushev M.V., Mazunin I.O., Vinogradova E.N., Kamenski P.A.
Mitochondrial fission and fusion
Biochemistry (Moscow) 2015

Patrushev M.V., Kamenski P.A., Mazunin I.O.
Mutations in mitochondrial DNA and approaches for their correction
Biochemistry (Moscow) 2014

Litvinova L.S., Kirienkova E.V., Mazunin I.O., Vasilenko M.A., Fattakhov N.S.
Pathogenesis of insulin resistance in metabolic obesity
Biochemistry (Moscow) Supplement Series B: Biomedical Chemistry 2014

Mazunin I.O., Volodko N.V., Starikovskaya E.B., Sukernik R.I.
Mitochondrial genome and human mitochondrial diseases
Molecular Biology 2010

Mazunin I.O.
Modern concepts of the mitochondrial structure and functions
Russian Journal of Genetics 2010

ORAL PRESENTATIONS

Preimplantation genetic testing for mitochondrial DNA mutations (PGT-mt).
X International Scientific and Practical Conference "MEETINGS ON THE VOLGA 2026: Current Topics in 21st-Century Clinical Embryology", Samara, July 23-25, 2026

Prevention of mitochondrial disease transmission: combined PGT-A/PGT-mt.
Genetico.Case Symposium, Moscow, July 15, 2026

Fetal gene therapy.
International Conference "Cornerstone Aspects of Reproductive Medicine XXI", St. Petersburg, May 28-31

Gene therapy for pathogenic variants in mitochondrial DNA.
VII All-Russian Multimedia Conference with International Participation "Gene Therapy: Present and Future", Moscow, May 13-14, 2026

Evolution of preimplantation genetic testing.
XI Scientific and Practical Conference "Current Issues in Embryology", Nizhny Novgorod, April 15-16, 2026

Oocyte rejuvenation technologies.
XI Scientific and Practical Conference "Current Issues in Embryology", Nizhny Novgorod, April 15-16, 2026

New eugenics: hype or our future?
International Conference "Cornerstone Aspects of Reproductive Medicine XX", Moscow, December 4-7, 2025

CRISPR-Cas12a as a tool for creating cellular and animal models of mitochondrial diseases.
I Congress of the National Society for Hematopoietic Stem Cell Transplantation, Gene and Cell Therapy, St. Petersburg, September 18-20, 2025

Improving oocyte quality after age 40: what can modern science offer?
International Conference "Cornerstone Aspects of Reproductive Medicine XIX", Moscow, June 5-8, 2025

The role of genetics in an embryologist's failures.
Conference "Current Issues in Embryology", Nizhny Novgorod, April 23-24, 2025

What should be added to culture medium to make life better for the embryo?
Conference "Current Issues in Embryology", Nizhny Novgorod, April 23-24, 2025

Genome editing technologies.
All-Russian Multimedia Conference "Gene Therapy: Present and Future", Moscow, April 7-8, 2025

Introduction of site-directed deletions into the human mitochondrial genome.
All-Russian Multimedia Conference "Gene Therapy: Present and Future", Moscow, April 7-8, 2025

PGT for mitochondrial DNA mutations.
Multidisciplinary Conference with International Participation "PGT-M: Flowers, Fruits and Stones", Moscow, March 22, 2025

Clinical prospects for correcting mutations in human mitochondrial DNA.
Scientific online webinar on gene therapy, Moscow, October 10, 2024

Diagnosis of mutations in mitochondrial DNA: from fertilization to human birth.
International Conference "Cornerstone Aspects of Reproductive Medicine XVII", Moscow, June 6-9, 2024

Molecular approaches to ranking embryos for transfer.
Conference "Current Issues in Embryology", Nizhny Novgorod, April 24-25, 2024

On the nature of mosaicism.
Conference "Current Issues in Embryology", Nizhny Novgorod, April 24-25, 2024

Genetics in human reproduction.
School of Reproductive Health in Ossetia, Vladikavkaz, April 19-20, 2024

Gene therapy for mitochondrial diseases.
All-Russian Multimedia Conference "Gene Therapy: Present and Future", Moscow, December 14, 2023

"Mitochondrial mathematics", or how to count cells in an embryo correctly.
International Conference "Cornerstone Aspects of Reproductive Medicine XVI", Moscow, December 7-10, 2023

Human mitochondrial genome surgery.
Meeting of the Young Scientists Club dedicated to genome editing, Moscow, November 16, 2023

Use of split guide RNA in human mitochondrial genome editing.
CRISPR-2023, Novosibirsk, September 11-13, 2023

Selection of human embryos for implantation based on mtDNA copy number and nucleotide sequence quality.
International Conference "Chromosome 2023", Novosibirsk, September 5-10, 2023

Mitochondrial genome as a predictor of embryo quality.
ART CLUB IVF, Sochi, March 3-4, 2023

The role of oocyte and embryo mitochondria in fertilization, development, and disease.
XXXII Annual International Conference of RAHR "Reproductive Technologies Today and Tomorrow", Kazan, September 7-10, 2022

Human mitochondrial genome surgery.
"Genome Editing in Medical Genetics 2021", Moscow, July 1-2, 2021

Causes and consequences of artificially created mtDNA heteroplasmy during nuclear transfer.
International Conference "Cornerstone Aspects of Reproductive Medicine XI", Moscow, December 3-6, 2020

Mitochondrial puzzles: attempts to see the whole picture.
International Conference "Cornerstone Aspects of Reproductive Medicine VIII", Moscow, December 5-8, 2019

Modern tools for genome surgery.
XXIX Annual International Conference of RAHR "Reproductive Technologies Today and Tomorrow", Rostov-on-Don, September 4-7, 2019

Genome editing: PRO ET CONTRA.
International Conference "Cornerstone Aspects of Reproductive Medicine VII", Moscow, June 6-9, 2019

Towards editing the human mitochondrial genome.
III International Conference "Science of the Future" and IV All-Russian Youth Scientific Forum "Science of the Future - Science of the Young", Sochi, May 14-17, 2019

Biological role of mitochondria. Technologies for manipulating mitochondrial DNA heteroplasmy levels.
International Conference "Cornerstone Aspects of Reproductive Medicine VI", Moscow, November 29 - December 2, 2018

Creation of cells with defined properties and functions.
Baltic Symposium on Immunology, Molecular and Regenerative Medicine, Kaliningrad, November 22-23, 2018

Technologies for removing mutant mtDNA as part of cytoplasmic replacement.
XXVIII Annual International Conference of RAHR "Reproductive Technologies Today and Tomorrow", Ufa, September 5-8, 2018

Mitochondrial Genome Surgery.
International Conference "Chromosome 2018", Novosibirsk, August 20-24, 2018

Hacking the human mitochondrial genetic code.
Mechanisms of Regulation of Organelle Functions in Eukaryotic Cells, Irkutsk, May 22-24, 2018 - plenary lecture (DOI:10.31255/978-5-94797-318-1-63-64)

Modern tools in genetic surgery.
Hybrid Technologies and Biomedical Applications, Kaliningrad, August 21-27, 2016

Towards editing organelle genomes.
Hybrid Technologies and Biomedical Applications, Kaliningrad, August 24-30, 2015

Mitochondrial genome in health and disease.
Scientific School on Molecular and Cellular Biology, Biotechnology and Molecular Aspects of Fundamental Medicine, Institute of Chemical Biology and Fundamental Medicine SB RAS, Novosibirsk, September 1-2, 2010

New data on the structure and functions of mitochondria.
International Conference "Chromosome 2009", Novosibirsk, August 31 - September 6, 2009

GRANTS

2024 competition "Basic and exploratory research by small individual research groups"; project "*Adaptation of the Argonaute protein system for targeting the human mitochondrial genome*" (2025-2026).

Skoltech Translational Research and Innovation Program (STRIP); project "*Development of mitochondrial therapy aimed at reducing aneuploidy rates in in vitro-matured oocytes*" (2024-2025).

2022 competition "Basic and exploratory research by international research teams" (BRFFI); project "*Optimization of the CRISPR-Cas system for efficient editing of the human mitochondrial genome*" (2023-2025).

2021 competition "Research by scientific groups led by young scientists" under the Presidential Research Projects Program implemented by leading scientists, including young scientists (RSF); project "*Mitochondrial genome dynamics in human embryonic development*" (2021-2024).

Russian Foundation for Basic Research, competition for interdisciplinary projects on "Fundamental problems of cell technologies"; project "*Development and application of technology for editing the CSBII region of mtDNA to study the mechanism controlling mitochondrial replication/transcription*" (2019-2022).

2017 competition "Research by scientific groups led by young scientists" under the Presidential Research Projects Program (RSF), implemented by leading scientists, including young scientists; project "*Molecular genetic technologies for targeted editing of human mitochondrial DNA*" (2017-2020).

Competitive selection of research projects carried out by research teams of research centers and/or scientific laboratories of higher education institutions (State Assignment); project "*Functional assessment of modified mitochondrial DNA resistant to deletion formation*" (2017-2019).

Russian Foundation for Basic Research; project "*Manipulation of mitochondrial DNA heteroplasmy levels in human cells using the CRISPR/Cas9 system*" (2015-2016).

Program "Research and Development in Priority Areas of the Scientific and Technological Complex of Russia for 2014-2020"; project "*Development of a gene therapy method for Leber hereditary optic neuropathy*" (2014-2016).

Program "Research and Development in Priority Areas of the Scientific and Technological Complex of Russia for 2014-2020"; project "*Development of a test system based on next-generation sequencing technology for molecular diagnosis of mitochondrial disorders*" (2014-2015).

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